

Bayesian Network Modeling of NFL Game Stats and Market Signals for Value Identification

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1 Problem Statement and Motivation

The problem we are trying to solve is: Does partially overlapping information sources encode independent predictive information? For this problem, the domain and data we've chosen is game stats and market signals for NFL games.

Our motivation is to uncover whether line movement (market signals) contains information beyond game stats that reveal whether market pricing is efficient or not. Full efficiency would mean that there are no market signals or game stats that encode information not encoded by both together. If additional information exists in game stats that is not encoded within market signals, then there is information that could be exploited to further move the line towards efficiency.

Our dataset spans 3,990 games over 15 seasons with 6 market data points per game including opening and closing spreads. The two data sources are joined by season and matchup (home and away teams). Our game stats data is from nflverse[1] and our market data is from SportsBookReviewsOnline.com[2]. Sports betting markets have changed substantially since our data window - particularly following the 2018 repeal of PASPA, which legalized sports betting across U.S. states and dramatically increased market participation. The patterns identified in this study reflect a specific historical period and are not intended as a prescriptive betting system.

Table 1: Sample rows from nfl_game_stats_raw (nflverse)

Season	Wk	Home	Away	H Score	A Score	H EPA/play	H TOs	A TOs
2007	1	IND	NO	41	10	0.055	1	3
2007	1	BUF	DEN	14	15	-0.074	1	1
2007	1	CLE	PIT	7	34	-0.331	5	1
2007	1	GB	PHI	16	13	-0.082	4	1
2007	1	HOU	KC	20	3	0.011	3	3

Table 2: Sample rows from sbr_nfl_odds_raw (SBR)

Season	Home	Visitor	Open Sprd	Close Sprd	Open Tot	Close Tot	ML Home	ML Away
2007-08	Indianapolis	NewOrleans	5.5	5.5	50.5	52.5	-240	200
2007-08	HoustonTexans	KansasCity	38.5	3.0	0.0	37.5	-200	170
2007-08	Buffalo	Denver	3.0	3.0	37.0	38.5	150	-170
2007-08	Cleveland	Pittsburgh	4.5	5.0	37.5	36.0	195	-235
2007-08	Jacksonville	Tennessee	6.5	7.5	37.5	39.5	-350	290

2 Approach

We plan to implement a Bayesian Network (BN) to test our hypothesis. The DAG structure shown in Figure 1 is a representation of our assumptions from the domain - historical stats drive the opening line, and the market reacts. Our

BN will formally test conditional independence, specifically the causal relationship between line movement and game outcome.

Our baseline will be Naive Bayes (NB) with assumption that all features are independent. We will use Hill Climbing as the main algorithm to encode our BN model, and we will use Peter-Clark (PC) as a validation check of the dependencies of our current DAG structure. We will test stats and market data in isolation and in combination. Comparing these three tests will tell us whether stats or market data is adding any signal or if they are redundant. If our BN can't beat Naive Bayes, then it means that the conditional dependencies we're modeling don't provide predictive value, which will formally confirm that closing lines are efficient.

Our dataset is dense with signal. First, the EPA per play is the most dense stat in football because it accounts for down, distance, and field position. Other stats, such as yards, turnovers, and scoring efficiency are highly correlated with EPA per play. We will also be able to derive a rolling average of Against-The-Spread (ATS) to substitute the actual ATS - because including the actual ATS would result in data leakage. Derivation of a rolling average ATS will be computed from previous games up to but not including the current game.

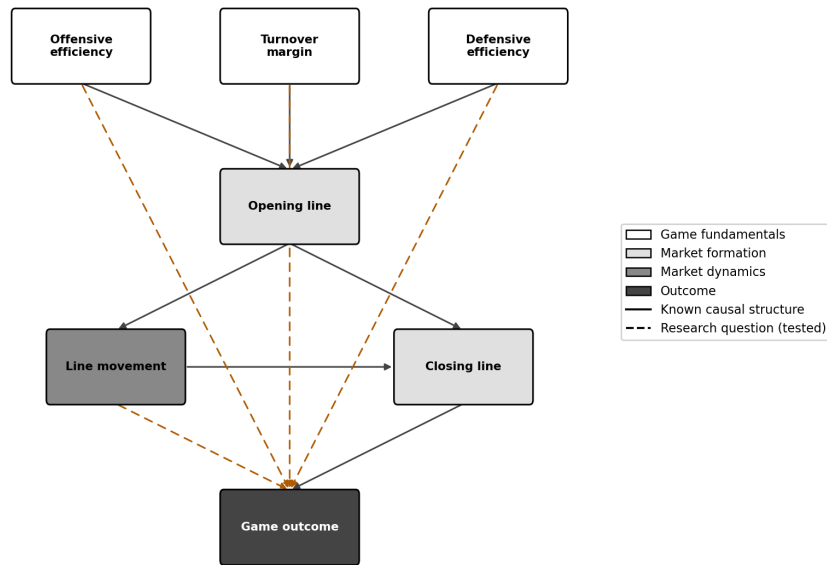


Figure 1: Bayesian network DAG structure. Dashed arrows represent hypothesized direct effects on game outcome — most critically, whether line movement predicts outcome independent of game fundamentals.

3 Related Work and Novelty

Prior work has shown that probabilistic graphical models can be useful for sports prediction. Constantinou, Fenton, and Neil developed Bayesian network models for association football forecasting, showing that BNs can represent uncertain relationships between team strength, form, subjective factors, and match outcomes [3], [4]. However, their models focus mainly on game-related variables rather than modeling betting market movement as its own source of information. Other work has combined historical performance data with bookmaker odds. Egidi, Pauli, and Torelli use a hierarchical Bayesian model that blends historical team data with betting odds [5], while Hubáček, Šourek, and Železný study machine-learning systems that compare model predictions against bookmaker odds [6]. These studies show that market information can be useful, but they do not model the conditional dependence structure between game statistics, opening lines, line movement, and outcomes using a Bayesian network.

Separate research has studied betting market efficiency and line movement directly. Simon analyzes real-time betting line movement and finds that betting markets are mostly reliable but not always efficient [7]. Moskowitz studies

sports betting as a market for detecting mispricing and momentum [8], while Szalkowski and Nelson examine NFL opening and closing lines as predictors of game outcomes [9]. Our project differs from these works by combining both perspectives: we use a Bayesian network to model game statistics and line movement together. The goal is not only to predict outcomes, but to test whether line movement contains additional predictive information after conditioning on game fundamentals. This makes the project a study of information overlap, conditional dependence, and market efficiency in NFL game pricing.

4 Evaluation Approach

For our three-class classification problem (cover / loss / push) with continuous features, we expect to start with the Gaussian Naive Bayes variant. We plan to compare stats-only, market-only, and combined models to test independent signal from the line movement. We plan to evaluate for predictive performance, calibration, and efficiency using Accuracy, Log Loss, and Brier Score. We may experiment with binary metrics, such as AUC-ROC, using a one-vs-rest approach.

5 Milestones

Table 3: Project Milestones

Due	Milestone	Owner
Apr 26	Data pipeline complete (acquisition + preprocessing)	Done
May 3	Project Proposal submitted	Done
May 7	Proposal Presentation	All
May 14	DAG validation via PC algorithm	TBD
May 17	Naive Bayes baseline implemented and evaluated	TBD
May 21	BN with Hill Climbing implemented	TBD
May 24	Ablation runs (stats only / market only / combined)	TBD
May 26	Results interpreted and written up	TBD
May 28	Paper draft complete	All
May 31	Final revisions and artifact submitted	All
Jun 4	Final Presentation	All

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